HL-50E Series - Troubleshooting Guide

TG0168

Revision B - 03/2026

Overview

This chapter provides a basic Troubleshooting guide for both the Haas Laser Marker and the Fume Extractor.

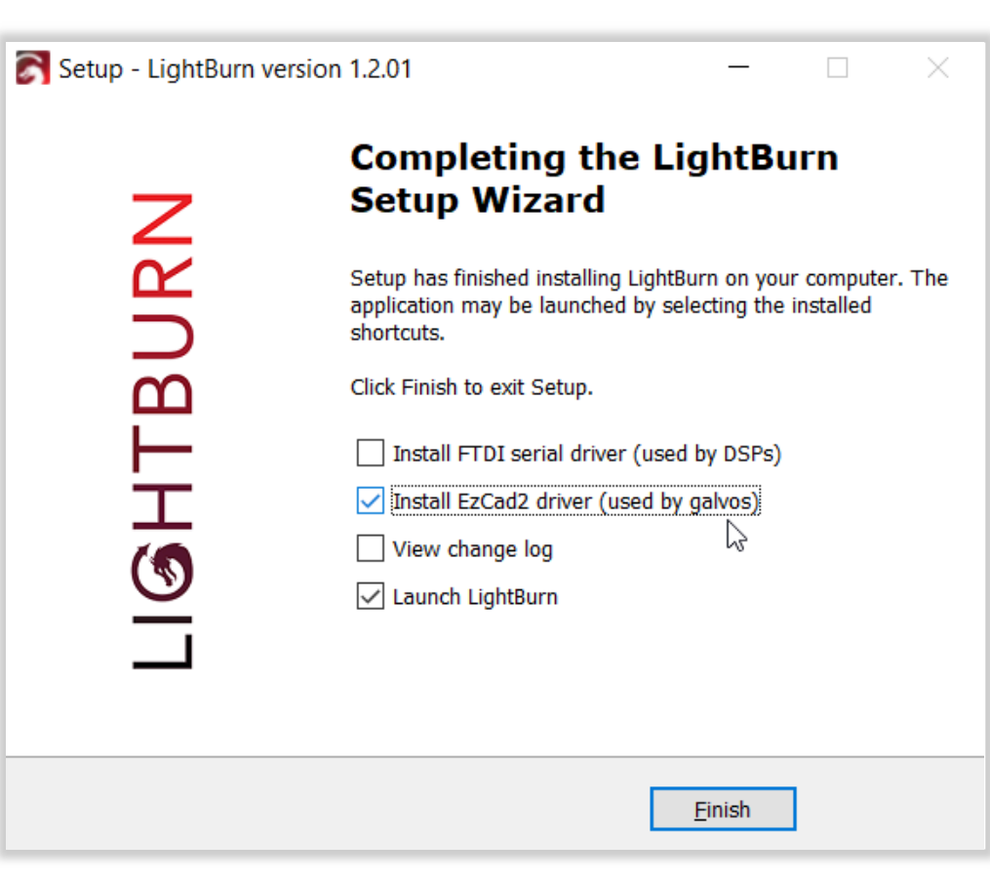
This guide includes a table of common issues with the laser marker and the fume extractor, as well as potential solutions.

Only an authorized Haas Factory Outlet distributor should service or repair a Haas machine that is protected by the original factory warranty. Servicing by any other party voids the warranty.

NOTE: If you cannot find the scenario that matches your situation, please contact service@haascnc.com.

HL-50E/XT Symptom Table

SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
"Please close door / check the door switch and try again" message appears when the door is closed.	Door safety sensor switch is obstructed, or positioned incorrectly.	Reposition the bracket to the sensor until the safety sensor switch is triggered by the door. Ensure there are no obstructions to the door safety sensor.
	Device settings for door sensor switch are not setup correctly.	See **Safety Sensor Door Switch** section below.
Laser won't turn on when "LASER POWER" is pressed.	Key to "LASER ENABLED" function is on the "OFF" position.	Ensure the key is set to the "ON" position, and the E-stop has been released. The "LASER POWER" button should be illuminated with a green LED when the machine is active.
The laser machine is active, but the LightBurn Laser Window displays a "Disconnected" message.	Laser marking control board is ejected (USBLMCV4) from the computer.	Disconnect and reconnect the USB cable from the laser marking control board to the PC, and check for other loose connctions. See the **Laser Connection** section below for more details.
	EZCAD2 drivers were not installed, or were installed improperly.	Uninstall and reinstall the necessary drivers for LightBurn. See the **Laser Drivers** section below for more details.
	LightBurn software is not up to date.	Update LightBurn to the latest released version. https://lightburnsoftware.com/pages/download-trial
	LightBurn window was closed during the middle of a job.	End all LightBurn tasks on task manager and reopen a .lbrn file. The Laser Window should display a "Ready" message.
When installing Lightburn, display message says "Server cannot be reached".	Drivers might not be up to date.	Update the laser drivers.
Skewed laser markings.	The axis and scale are inverted and/or are bulging on the material.	Lens needs to be calibrated. Refer to Galvo Lens Calibration.
Laser head does not move up or down when commanded with buttons.	Coupler from motor to ballscrew is loose.	Remove laser head column cover to access the coupler and tighten the coupler set screw from the motor side.
Laser head is rough to move with handle and/or with the up and down buttons.	Screw handle hard stop is tightened	Loosen the screw handle. See the **Laser Head Up and Down** section.
	Lead screw may be binding or bowed.	Remove the cover to access the lead screw. Loosen the coupler and manually hand jog the laser head and check for binding or bowing of the lead screw.
Foot switch pedal fails to Start or Stop Laser operation.	Foot switch pedal parameters may be misconfigured in device settings.	Adjust parameters to enable the foot switch pedal. See **Foot Switch Pedal** section below.
"Boot failure detected" display on the monitor screen on initial power up.	Loose PC connections to the SSD hardrive.	Remove the PC and acess once of the panels to check that the connection the SSD card are pushed all the way in.
	If the SSD failed after checking for loose connection to the SSD card, it could be a bad SSD card.	Replace the SSD.
Laser fails to frame and attempts to engrave, but only produces a stationary laser dot moving straight downward with little to no visible marking on the material.	15VDC Power Supply is not functioning.	Check to see if the 15VDC power supply has the green LED ON and is reading the following voltages. - **GND to +15VDC = +15VDC** - **GND to -15VDC- = -15 VDC** - **+15VDC to -15VDC = 30 VDC** If 15VDC Power Supply fails the troubleshooting, contact **haasservice@haascnc.com** for further assistance. Include a comprehensive summary of the diagnostic steps performed, test results obtained, and all relevant findings to facilitate efficient evaluation and resolution of the issue.
Laser frames but does not mark.	Door Safety Sensor Switch is not being triggered.	See the **Door Safety Sensor Switch - Laser Not Marking** section.
	Laser Drivers are not installed.	See the **Laser Drivers** section.
	24VDC Power Supply is not functioning.	Check to see if the 24V power supply has the green LED ON, the cooling fan on the 24V power supply is running when laser is turned on, and is reading the following voltages: - **GND to Neutral = 0 Vdc** - **GND to V- = 0 Vdc** - **GND to V+ = 24 Vdc** - **V- to V+ = 24 Vdc.** If 24V Power Supply fails the troubleshooting, the 24V Power Supply is nonfunctional and will have to be replaced by ordering **93-4715** 24V POWER SUPPLY HL-50E/XT.
	Laser Source Failure	If all prior troubleshooting procedures have been completed and the issue persists, contact **haasservice@haascnc.com** for further assistance. Include a comprehensive summary of the diagnostic steps performed, test results obtained, and all relevant findings to facilitate efficient evaluation and resolution of the issue.



Setup - LightBurn version 1.2.01

Completing the LightBurn Setup Wizard

Setup has finished installing LightBurn on your computer. The application may be launched by selecting the installed shortcuts.

Click Finish to exit Setup.

☐ Install FTDI serial driver (used by DSPs)

☑ Install EzCad2 driver (used by galvos)

☐ View change log

☑ Launch LightBurn

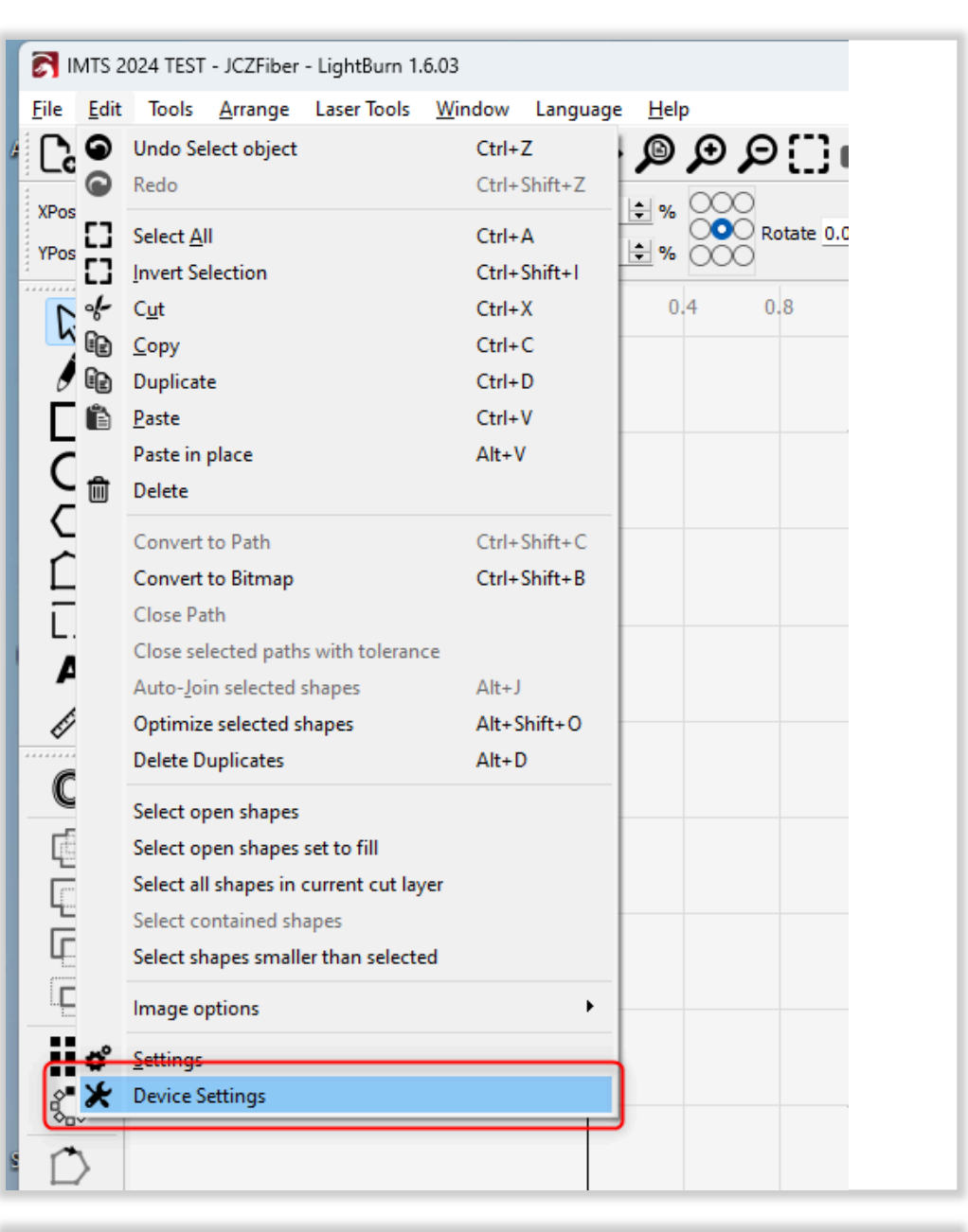
LIGHTBURN
Finish

NOTE: EZCAD2 LMC drivers are required to run LightBurn laser control software. The EZCAD2 LMC drivers and EZCAD2 laser control software are two separate entities. **Do NOT download the EZCAD2 laser control software if you intend to use LightBurn (preferred).**

The EZCAD2 LMC drivers can be installed upon updating or installing LightBurn, as shown in the image on the left. The LightBurn setup wizard can also be found in the following link: HTTPS://LIGHTBURNSOFTWARE.COM/PAGES/DOWNLOAD-TRIAL

Alternately, the installer for the EZCAD2 LMC drivers is in the LightBurn files at "C:\Program Files\LightBurn\EzCad2Driver\" and can be run again via the "dpinst-x64.exe" file directly.

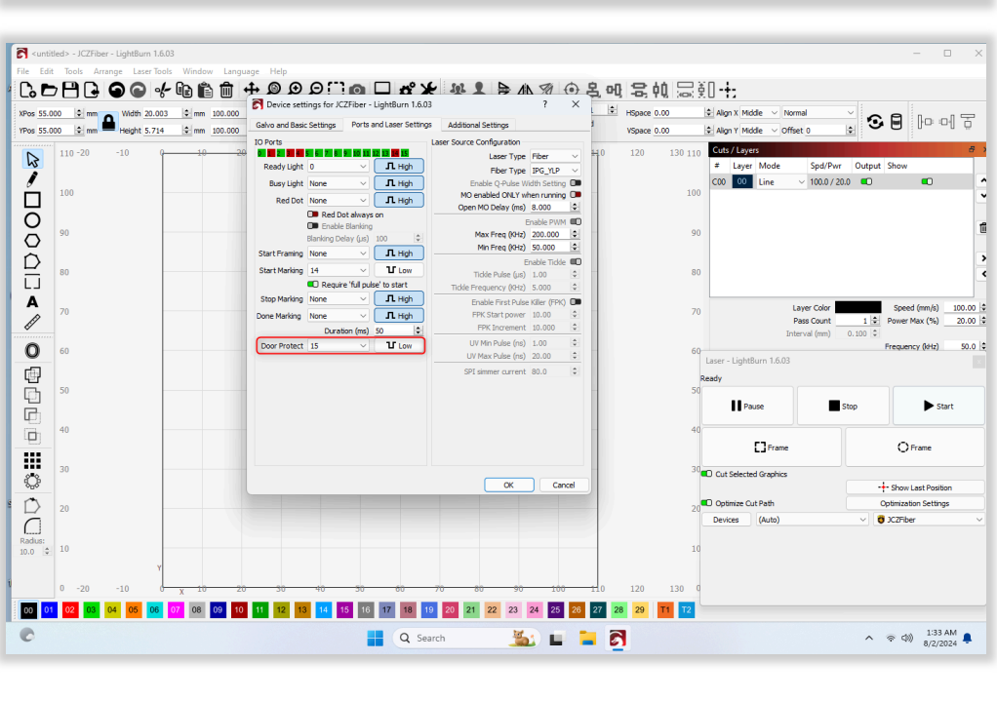
Door Safety Sensor Switch



IMTS 2024 TEST - JCZFiber - LightBurn 1.6.03
File Edit Tools Arrange Laser Tools Window Language Help
Undo Select object Ctrl+Z
Redo Ctrl+Shift+Z
Select All Ctrl+A
Invert Selection Ctrl+Shift+I
Cut Ctrl+X
Copy Ctrl+C
Duplicate Ctrl+D
Paste Ctrl+V
Paste in place Alt+V
Delete
Convert to Path Ctrl+Shift+C
Convert to Bitmap Ctrl+Shift+B
Close Path
Close selected paths with tolerance
Auto-Join selected shapes Alt+J
Optimize selected shapes Alt+Shift+O
Delete Duplicates Alt+D
Select open shapes
Select open shapes set to fill
Select all shapes in current cut layer
Select contained shapes
Select shapes smaller than selected
Image options
Settings
Device Settings

1

Go to **EDIT** > **DEVICE SETTINGS** to get to the Ports and Laser Settings.



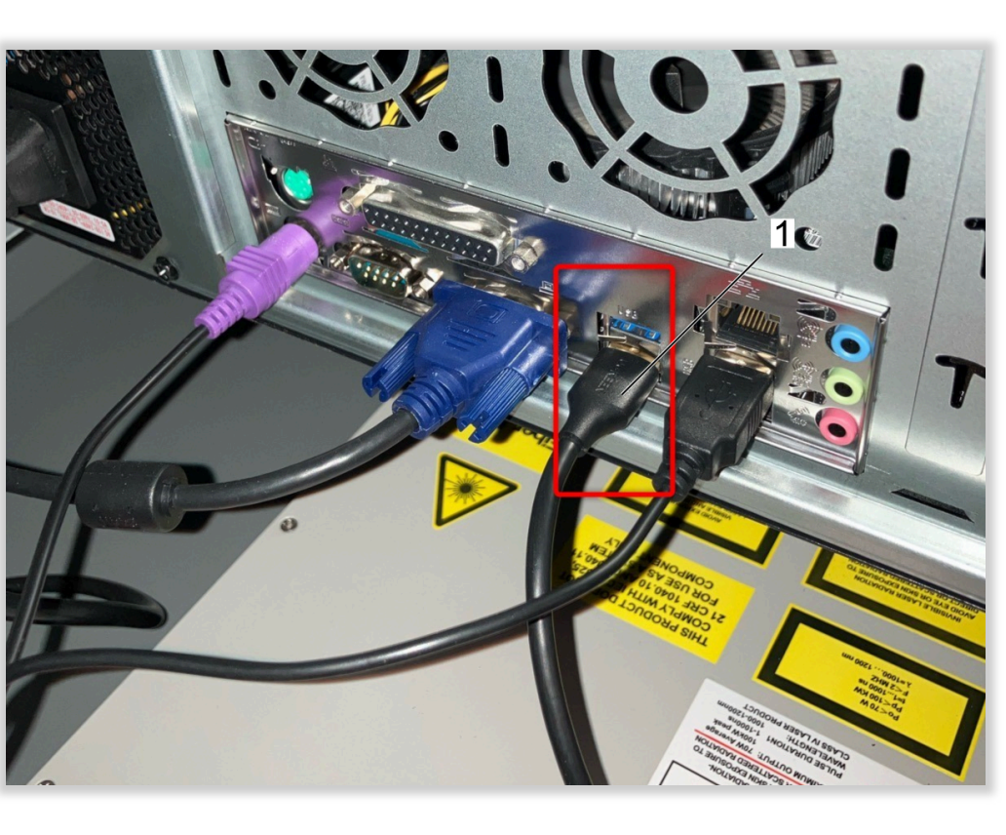
2

In the **Ports and Laser Settings**, the settings for Safety Sensor Door Switch should match to what is being shown in the image for **Door Protect**.

Verify by marking a test piece to see if the message appears when trying to mark with the door slightly open.

CAUTION: Be sure to use laser safe protective eyewear when marking.

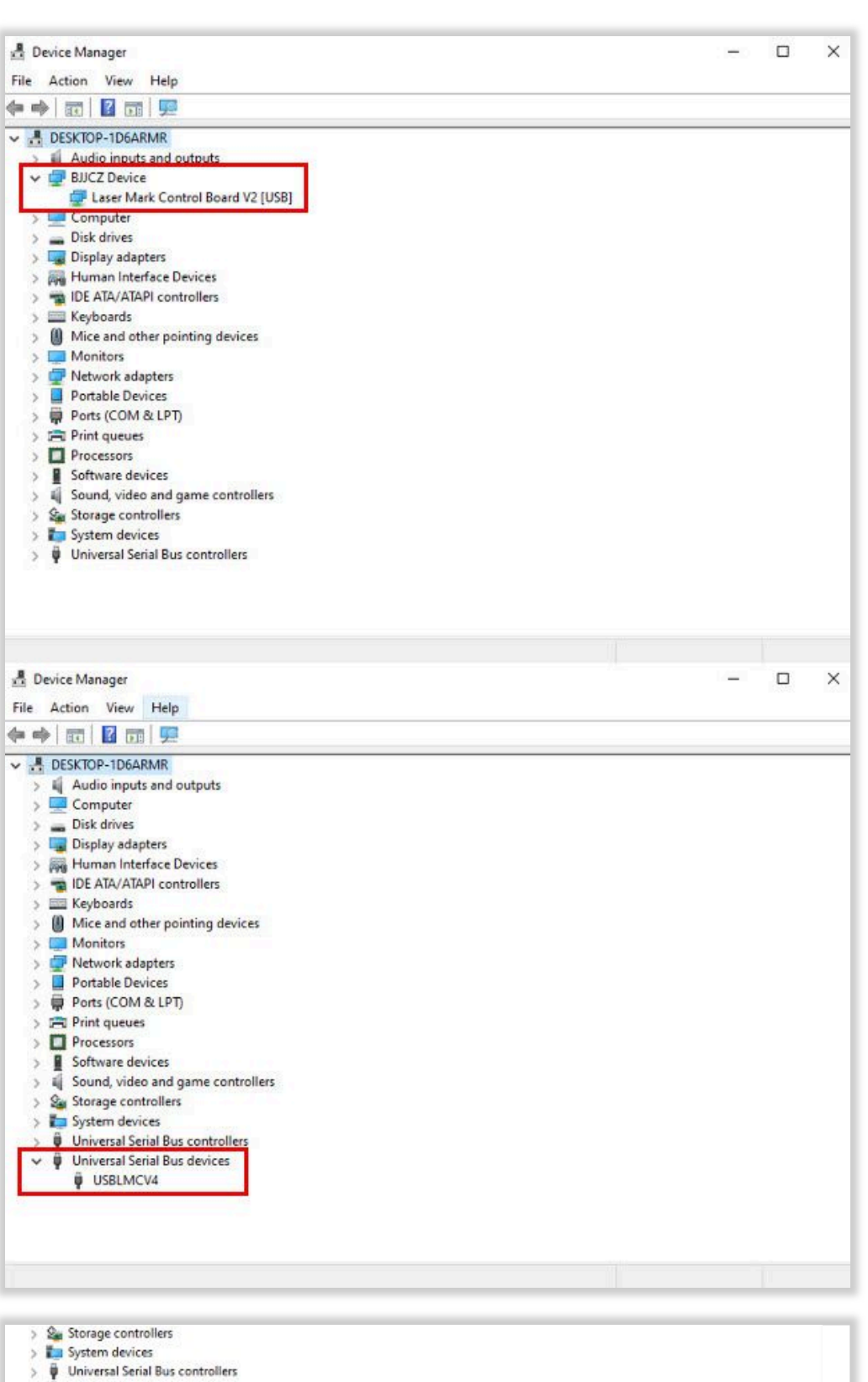
Laser Connection



Unplug and plug back in the laser usb cable [1] in the back of the PC as shown in the image.

Check to see if it appears in the device manager under "Universal Serial Bus devices".

Laser Drivers



Device Manager
DESKTOP-1D6ARMR
BJJCZ Device
Laser Mark Control Board V2 [USB]
Universal Serial Bus devices
USBLMCV4

1

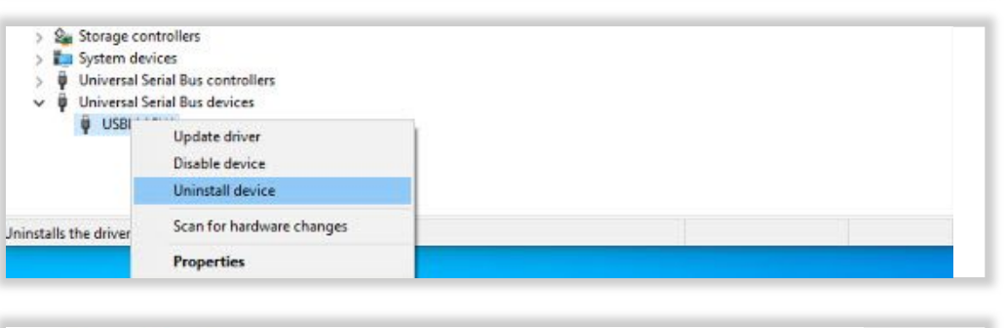
Begin by opening the Windows Device Manager in the machine's PC.

If the drivers were never installed, the PC will display a "BJJCZ Device" in the Window Device Manager.

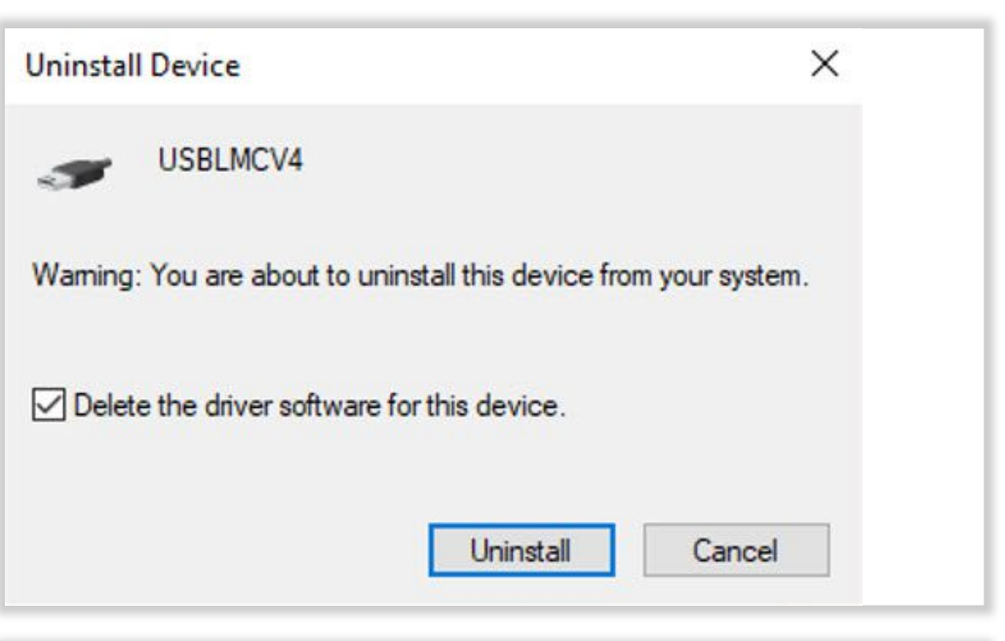
NOTE: Laser should be on so that the PC can display the device under "Universal Serial Bus devices" tree if it is installed properly.

If the drivers were installed properly, the PC will display a generic "USBLMCV2", a "USBLMCV4", or something similar under the "Universal Serial Bus devices" tree.

If neither of these devices appear in the Windows Device Manager, skip to **Step 3**.



Update driver
Disable device
Uninstall device
Scan for hardware changes
Properties



Uninstall Device

USBLMCV4

Warning: You are about to uninstall this device from your system.

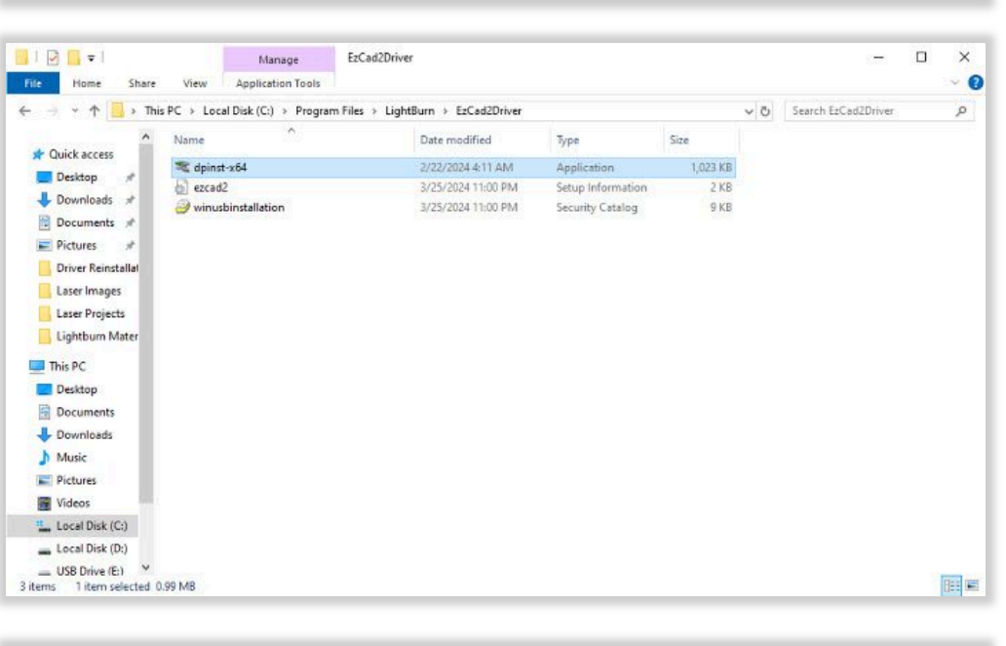
☑ Delete the driver software for this device.

Uninstall
Cancel

2

In either scenario, right-click your respective device and choose 'Uninstall Device' from the pop-up menu.

In the next window, check the box to delete the driver. This completely disassociates the driver from your device. Click the 'Uninstall' button to complete the process. When the process is complete, restart the PC.



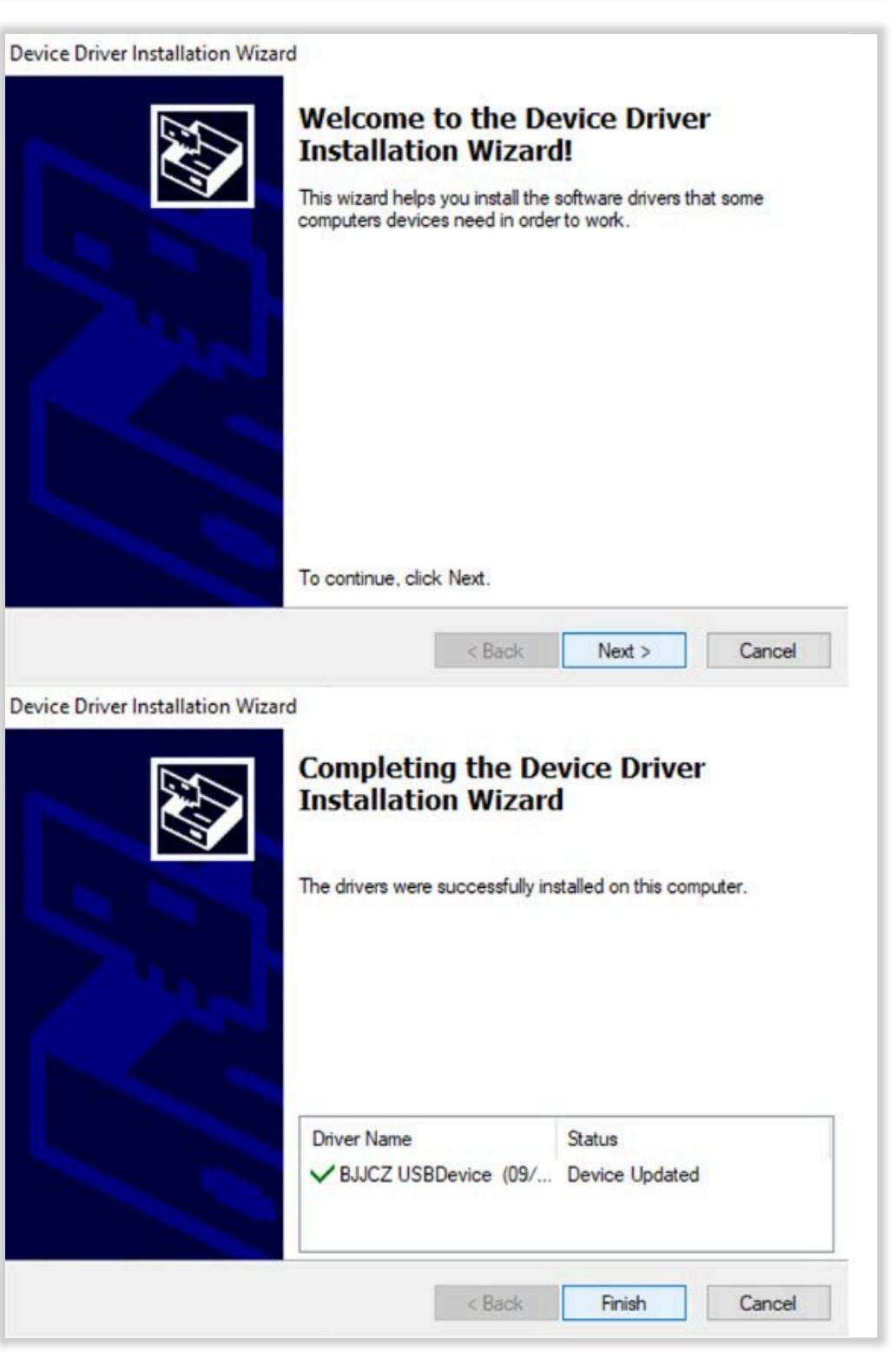
This PC › Local Disk (C:) › Program Files › LightBurn › EzCad2Driver
dpinst-x64
ezcad2
winusbinstallation

3

Navigate to C:\Program Files\LightBurn\EzCad2Driver\ and run the included setup file (dpinst-x64.exe) to reinstall the necessary drivers for LightBurn. When the setup installation wizard is complete, restart the PC.

Once the PC has been restarted, reopen LightBurn and ensure the Laser Window displays a 'Ready' message.

Verify that the Windows Device Manager displays a new instance of "USBLMCV2", "USBLMCV4", or something similar under the "Universal Serial Bus devices" tree.



Device Driver Installation Wizard

Welcome to the Device Driver Installation Wizard!

This wizard helps you install the software drivers that some computers devices need in order to work.

To continue, click Next.

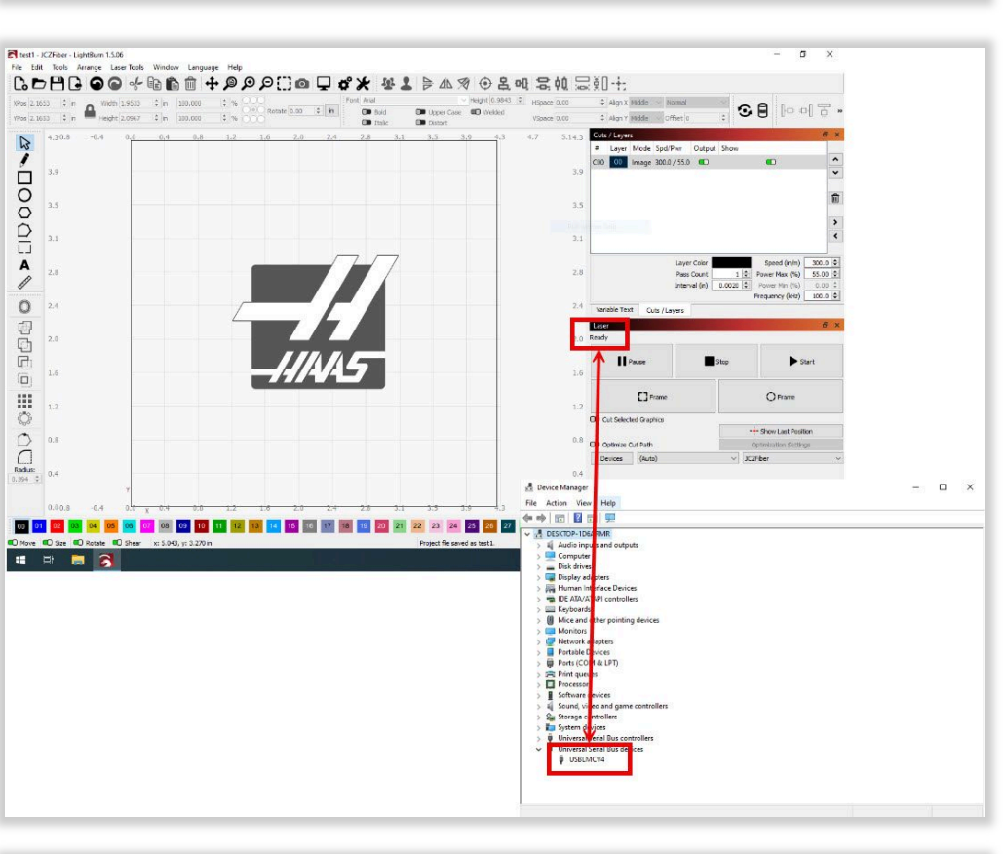
Device Driver Installation Wizard

Completing the Device Driver Installation Wizard

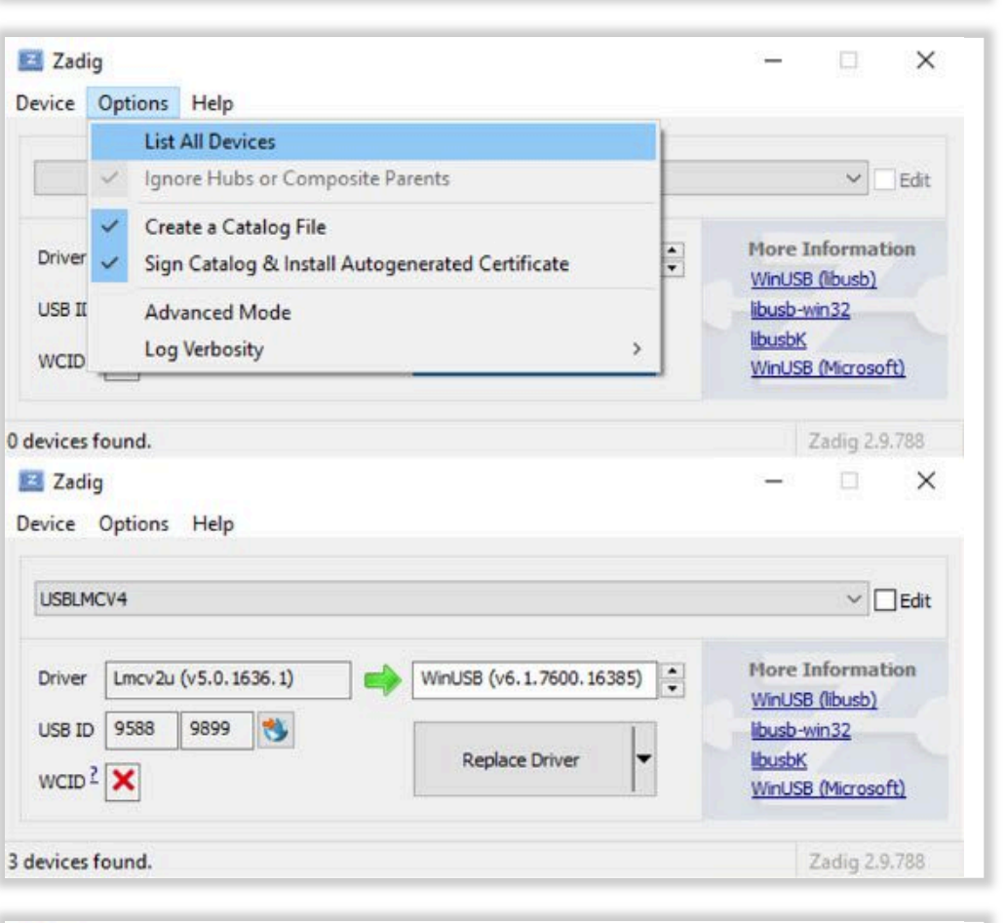
The drivers were successfully installed on this computer.

Driver Name	Status
BJJCZ USBDevice (09/...	Device Updated

< Back
Next >
Cancel
Finish



Ready
USBLMCV4



Zadig
Device Options Help
List All Devices
Ignore Hubs or Composite Parents
Create a Catalog File
Sign Catalog & Install Autogenerated Certificate
Advanced Mode
Log Verbosity
More Information
WinUSB (libusb)
libusb-win32
libusbK
WinUSB (Microsoft)
Replace Driver
Zadig 2.9.788

0 devices found.

USBLMCV4

Driver Lmcv2u (v5.0.1636.1) → WinUSB (v6.1.7600.16385)

USB ID 9588 9899

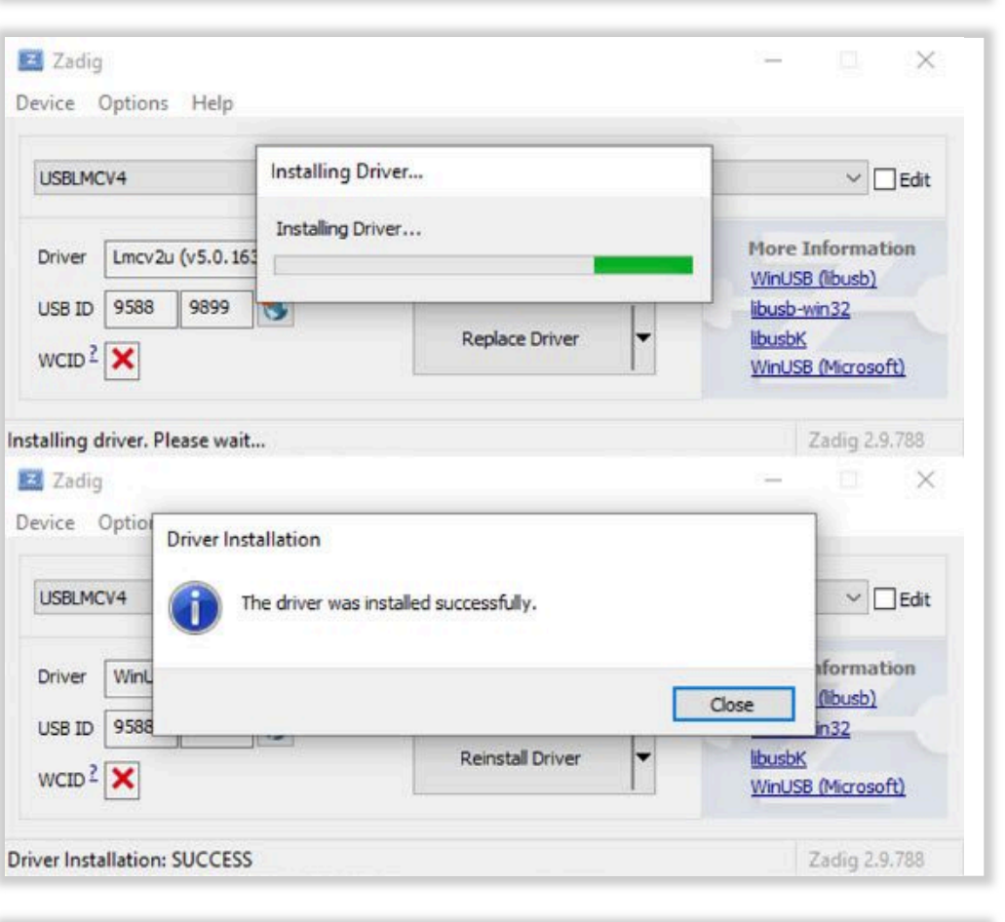
3 devices found.

4

In the event the driver swap performed by running the installer isn't working, try using Zadig (a driver tool) to replace the manufacturer's with a generic driver so LightBurn can properly communicate.

Download Zadig, the driver replacement tool from HTTPS://ZADIG.AKEO.IE/ and run it (does not require additional installation).

With the Zadig window open, open the 'Options' menu and enable the 'List All Devices' setting. In the primary device list drop-down, select 'USBLMCV2' or 'USBLMCV2' (or similar). WinUSB should be the target driver.



Installing Driver...

Installing driver. Please wait...

Driver Installation

The driver was installed successfully.

Driver Installation: SUCCESS

Replace Driver
Reinstall Driver
Close
Zadig 2.9.788

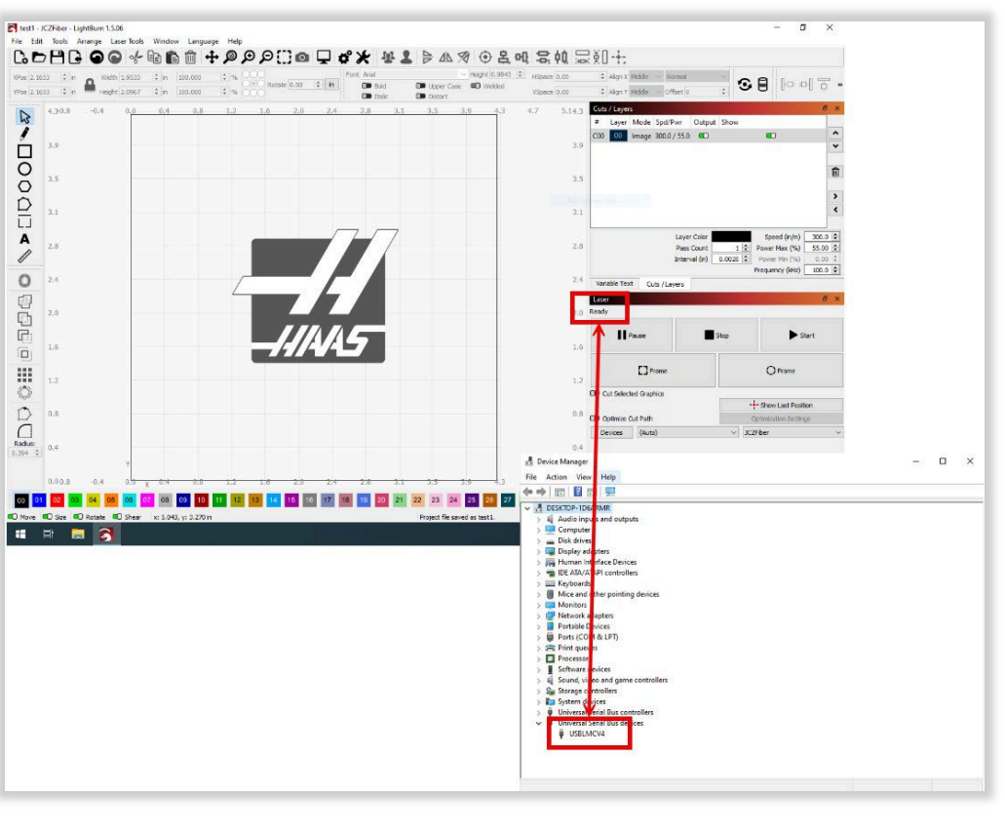
5

Click 'Replace Driver' and wait for the process to complete. On the rare occasion that Zadig fails, try running the function again.

When the Zadig driver installation wizard is complete, restart the PC.

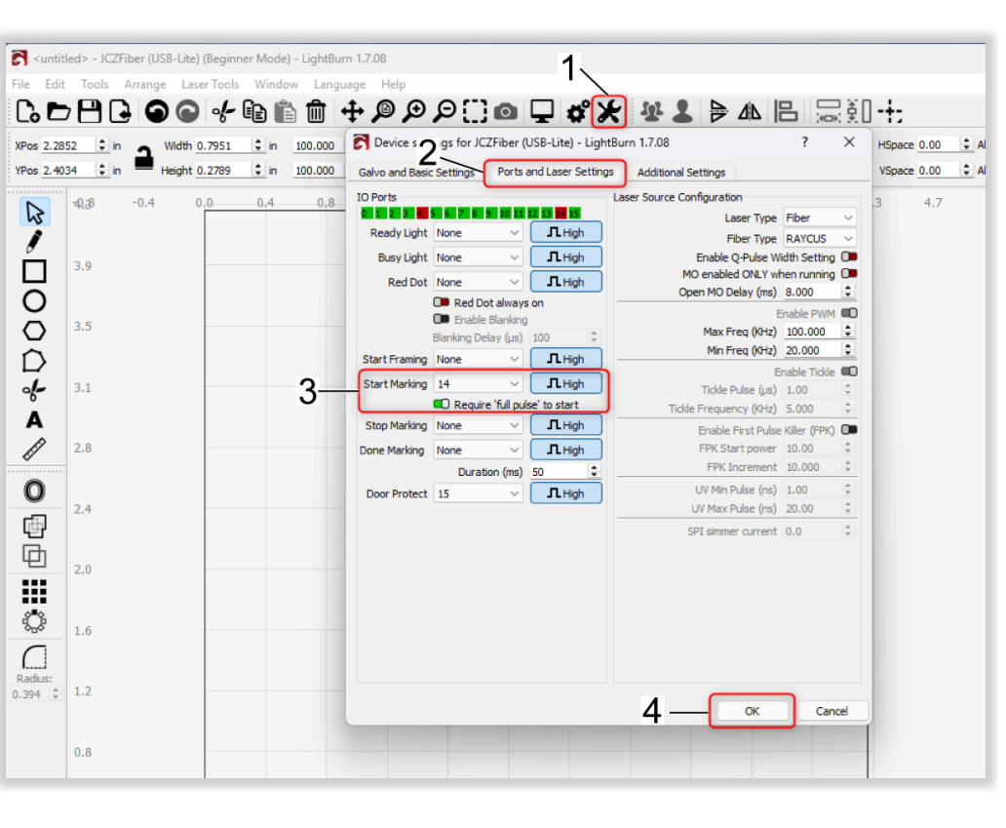
Once the PC has been restarted, reopen LightBurn and ensure the Laser Window displays a 'Ready' message.

Verify that the Windows Device Manager displays a new instance of "USBLMCV2", "USBLMCV4", or something similar under the "Universal Serial Bus devices" tree.



Ready
USBLMCV4

Foot Switch Pedal



1
2
3
4
Ports and Laser Settings
Start Marking 14
High
Require 'full pulse' to start
OK

1

Go to Device Setting [1].

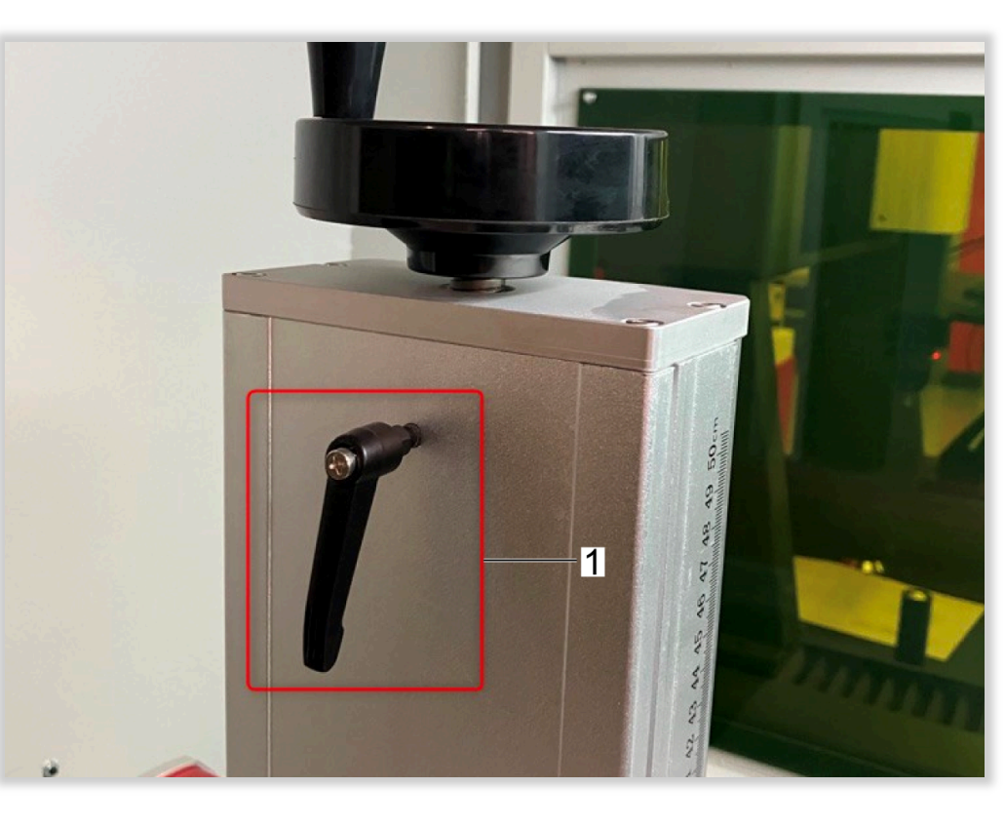
Select Ports and Laser Settings [2].

Adjust the Start Marking setting to 14 and High, and that the "full pulse" is set to ON [3].

Click OK [4].

Run a test to check if pressing the foot switch pedal initiates framing, and press it again to begin marking.

Laser Up and Down

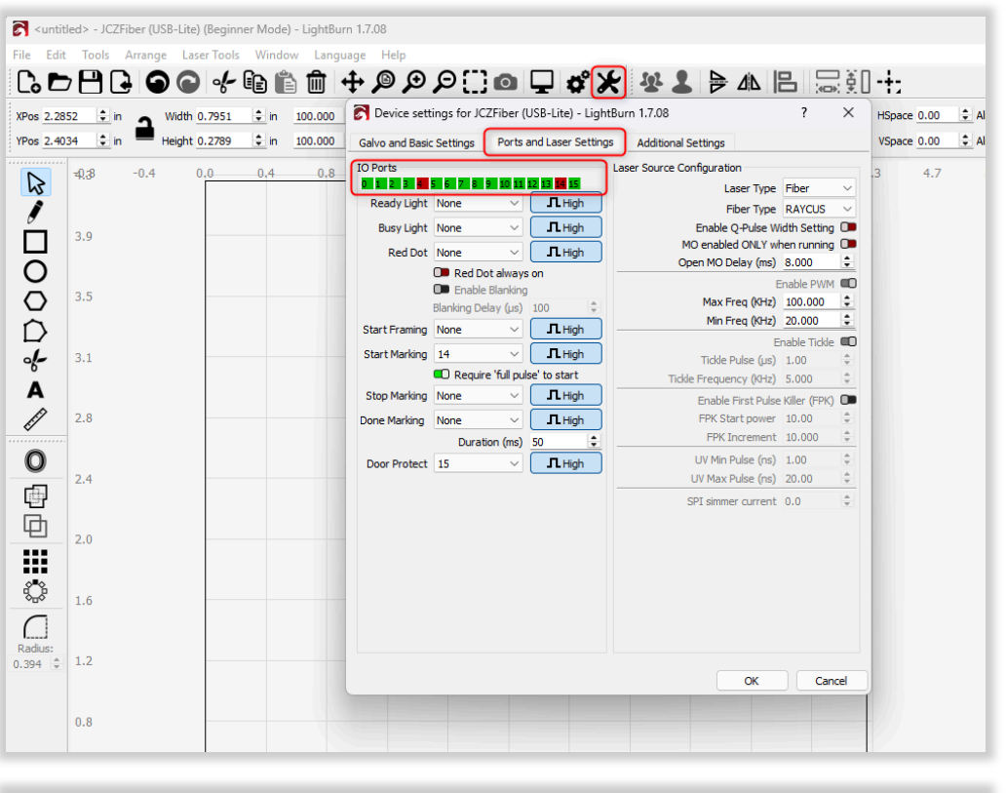


1

Loosen the Handle Screw [1].

Manually jog the head up and down to see if that allows the laser head to move up and down smoothly.

Door Safety Sensor Switch - Laser Not Marking



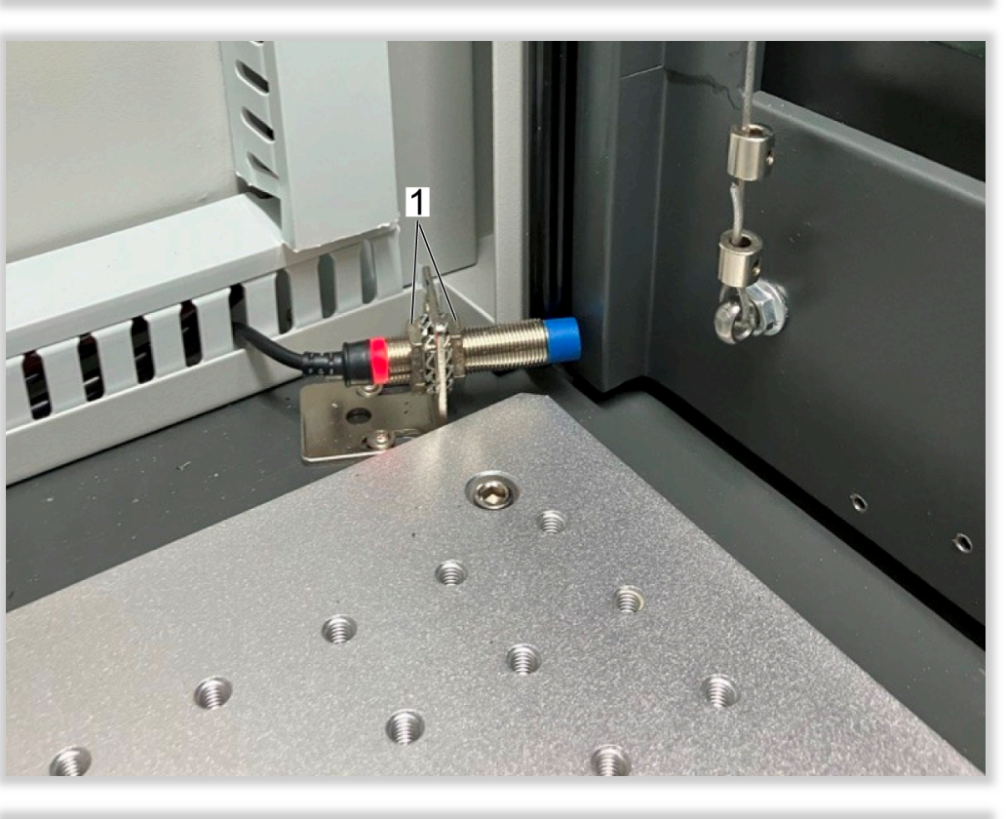
IO Ports
Ports and Laser Settings
Door Protect 15

1

In LightBurn, access the **Device Settings** panel by selecting the **Wrench** icon. Navigate to the **Ports and Laser Settings** section, then locate the **I/O Ports** list.

Ensure the laser system is powered on but not actively marking. Open and close the machine door while observing **Port 15** in the I/O display. Verify whether the port state changes when the door is actuated.

If **Port 15** does not change state during this test, proceed to the next troubleshooting step.

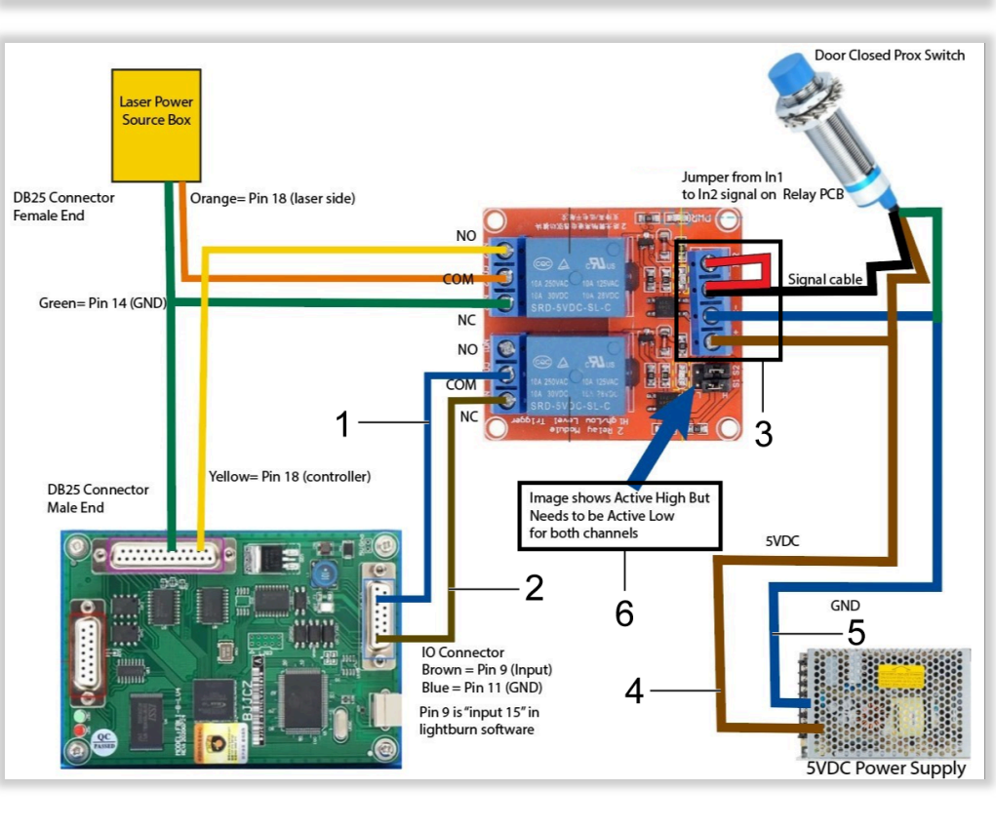


2

If the door safety sensor switch is not being actuated by the door, manually test the switch using a metal object to simulate activation.

If the switch responds when triggered manually, adjust its position. Loosen the retaining nuts on both sides [1], then reposition the switch by moving it inward or outward as required. Secure the nuts once the switch is properly aligned so that normal door movement reliably actuates the sensor during opening and closing.

If the door safety sensor switch does not respond during manual activation, proceed to the next troubleshooting step.



Laser Power Source Box
DB25 Connector Female End
Orange= Pin 18 (laser side)
Green= Pin 14 (GND)
Yellow= Pin 18 (controller)
DB25 Connector Male End
NO
COM
NC
NO
COM
NC
Jumper from In1 to In2 signal on Relay PCB
Signal cable
Door Closed Prox Switch
Image shows Active High But Needs to be Active Low for both channels
5VDC
GND
IO Connector
Brown = Pin 9 (Input)
Blue = Pin 11 (GND)
Pin 9 is "input 15" in lightburn software
5VDC Power Supply
1
2
3
4
5
6

3

If the door safety sensor switch fails to operate, verify that the **5 VDC power supply** is functioning correctly. Measure the output voltage using a calibrated multimeter to confirm proper voltage is present. Additionally, inspect all wiring connections to the relay to ensure they are secure and properly terminated.

Inspect the following relay connections:

- **[1] COM cable** on the dual-channel relay
- **[2] NC cable** on the dual-channel relay
- **[3]** Wiring to **IN1, IN2, DC−, and DC+** terminals on the relay PCB
- **[4] 5 VDC supply cable** connected to the 5 VDC power supply and to **DC+** on the dual-channel relay
- **[5] GND connection** from the 5 VDC power supply to **DC−** on the dual-channel relay
- **[6]** Confirm that the **Level Trigger** setting is configured to **Low** for both relay channels

If the door safety sensor switch continues to fail after verifying voltage and confirming all relay connections, replace the component with part number **93-5200 - Door Safety Sensor Switch HL-50E/XT**.

Troubleshooting - Fume Extractor

SYMPTOM	POSSIBLE CAUSE	CORRECTIVE ACTION
Fume Extractor is not functioning when switched on.	Knob is set to "Off"	Turn the knob clockwise to set the intensity of the vacuum.

HL-50E Series - Electrical Diagrams

Revision A - 09/2025

Introduction

This procedure outlines the electrical schematics associated with the HL-50E/XT unit. It serves as a reference for technicians involved in troubleshooting.

HL-50E Wiring Schematics

NOTE: The electrical schematics for the HL-50E can be used as a reference for the HL-50E-XT, as the only difference between electrical components in the two models is the layout of components within the electrical cabinet.

The electrical schematics for the HL-50E are available at the following link:

HL-50E ELECTRICAL DIAGRAMS

Feedback